

## §4.2

# Prime and Composite Numbers

---

Factors, primes, and prime factorization

# Today's Goal

---

After completing this section, you should be able to:

- Use rectangular arrays of squares to find all of the factors of a whole number
- Classify a whole number as prime, composite, or neither, by counting its distinct factors
- Find the prime factorization of a whole number using a factor tree, and explain why two different trees for the same number give the same prime factorization

# Today's Goal, Continued

---

- Decide whether a number is prime by testing only the prime divisors up to its square root
- Use the Sieve of Eratosthenes to produce the list of prime numbers up to a given number

# Learning Target 11

---

## LT.11 Prime and Composite Numbers

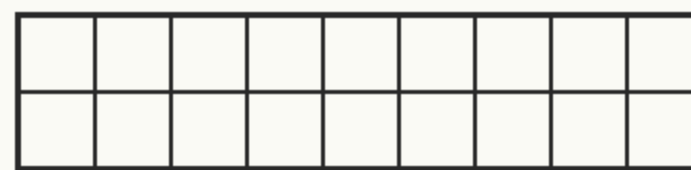
I understand what prime and composite numbers are and can determine whether a whole number is prime or composite (or neither). I can also find the prime factorization of a whole number.

# Rectangular Arrays of 18 Squares

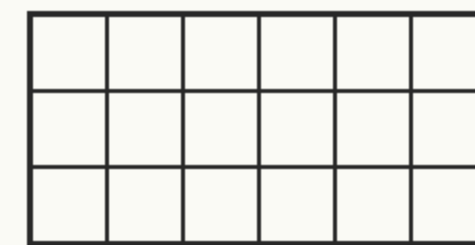
How many different rectangular arrays can you make with 18 squares?



$$1 \times 18$$



$$2 \times 9$$



$$3 \times 6$$

So the factors of 18 are 1, 2, 3, 6, 9, and 18.

## Definition 4.2.1

A whole number  $d$  is called a **factor** (or **divisor**) of another whole number  $n$  if  $n$  is divisible by  $d$ .

# Prime and Composite Numbers

---

With your group, use number cubes to build the rectangular arrays for each whole number from 2 to 12. Which numbers make only one rectangular array?

2, 3, 5, 7, and 11: each one has exactly two factors, 1 and itself.

# Prime and Composite Numbers

---

## Definition 4.2.4

If a whole number  $p$  has exactly two *distinct* (whole number) factors, then  $p$  is called a **prime number**.

## Definition 4.2.5

If a whole number  $n$  has more than two *distinct* (whole number) factors, then  $n$  is called a **composite number**.

# Three Observations

---

## Note 4.2.6

There are three important observations regarding the prime and composite numbers.

1. The number 1 is not a prime number because it has only one distinct factor: 1.
2. The number 1 is not a composite number because it does not have more than two distinct whole number factors.

# Three Observations, Continued

---

## Note 4.2.6, continued

3. The number 2 is the only even number that is a prime number. Every other even number  $n$  will have at least three distinct factors: 1, 2, and  $n$ .

# Prime Factorization

---

## Theorem 4.2.7

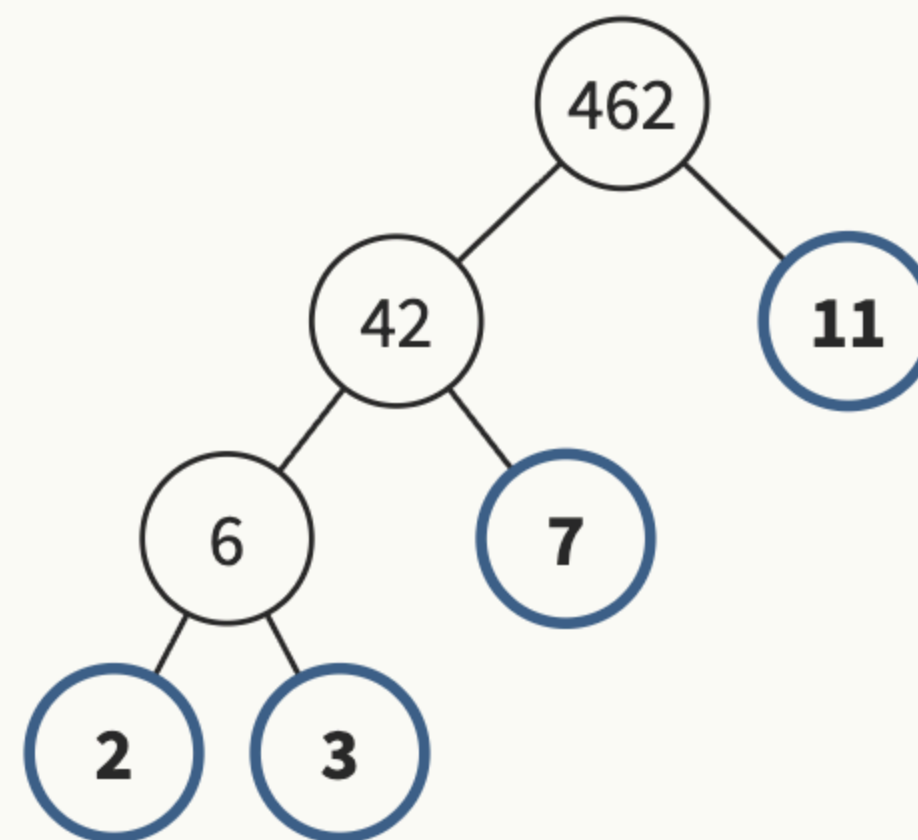
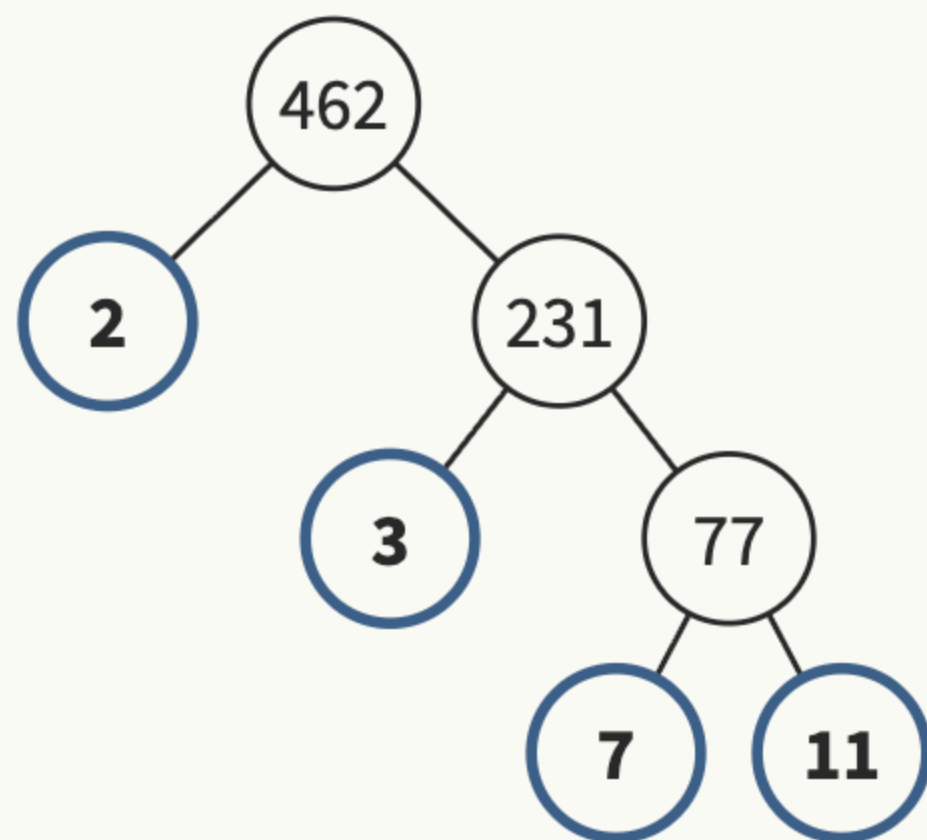
Every whole number greater than 1 is either prime or composite.

## Theorem 4.2.8 Fundamental Theorem of Arithmetic

Every whole number greater than 1 can be written as a *unique* product of prime factors, up to reordering the factors in the product.

# Prime Factorization of 462

Find the prime factorization of 462.



Both factor trees give us the prime factorization of 462:  $2 \times 3 \times 7 \times 11$ .

# Your Turn

---

**(a)** Is 51 prime, composite, or neither?

The digits of 51 add up to 6, so 51 is divisible by 3. Since  $51 = 3 \times 17$ , it is composite.

**(b)** Find the prime factorization of 360. Compare your factor tree with the trees in your group.

$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$ , whichever tree you drew.

# Factor Pairs of 100

---

List the pairs of factors that multiply to 100.

$$1 \times 100 = 100$$

$$20 \times 5 = 100$$

$$2 \times 50 = 100$$

$$25 \times 4 = 100$$

$$4 \times 25 = 100$$

$$50 \times 2 = 100$$

$$5 \times 20 = 100$$

$$100 \times 1 = 100$$

$$10 \times 10 = 100$$

After the first factor reaches 10, the remaining pairs of factors are just duplicates of the previous pairs with the factors switched. So it is sufficient to check for divisibility by prime numbers up to the square root of  $n$ .

# Is 527 Prime?

---

Is 527 a prime number? Since  $\sqrt{527} \approx 22.96$ , we only need to check the primes 2, 3, 5, 7, 11, 13, 17, and 19.

- 2 and 5: the ones digit is 7, which is not even and not 0 or 5.
- 3: the digits add up to  $5 + 2 + 7 = 14$ , which is not divisible by 3.
- 7:  $52 - 2(7) = 38$ , which is not divisible by 7.
- 11:  $5 - 2 + 7 = 10$ , which is not divisible by 11.

# Is 527 Prime? Continued

---

For 13, 17, and 19 we do not have any divisibility tests. So we will just check by dividing.

- 13:  $527 \div 13 = 40 \text{ R } 7$ .
- 17:  $527 \div 17 = 31 \text{ R } 0$ .

So  $527 = 17 \times 31$ , and 527 is composite.

# Is 221 Prime? Is 223 Prime?

---

Is 221 a prime number? Is 223 a prime number?

Since  $\sqrt{221} \approx 14.9$  and  $\sqrt{223} \approx 14.9$ , we only need to check 2, 3, 5, 7, 11, and 13.

Neither number is divisible by 2, 3, 5, 7, or 11.

$221 \div 13 = 17$ , so  $221 = 13 \times 17$  is composite.

$223 \div 13 = 17 \text{ R } 2$ , so 223 is prime.

# The Sieve of Eratosthenes

---

## Procedure 4.2.12 Sieve of Eratosthenes

1. Given a number  $n$ , we list of consecutive numbers from 2 to  $n$ .
2. We know that 2 is prime. Cross out all multiples of 2 from the list since 2 will be a factor of those numbers.
3. The next number in the list that has not been eliminated must be prime.

# The Sieve of Eratosthenes, Continued

---

## Procedure 4.2.12, continued

4. Cross out all multiples of the new prime number from the list since the new prime number will be a factor of those numbers.
5. Repeat Steps 3-4 until there are no remaining numbers on the list other than the prime numbers.

# Sieve Up to 60

Use the Sieve of Eratosthenes to find all prime numbers up to 60.

|               |               |               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 1             | <b>2</b>      | <b>3</b>      | <del>4</del>  | <b>5</b>      | <del>6</del>  | <b>7</b>      | <del>8</del>  | <del>9</del>  | <del>10</del> |
| <b>11</b>     | <del>12</del> | <b>13</b>     | <del>14</del> | <del>15</del> | <del>16</del> | <b>17</b>     | <del>18</del> | <b>19</b>     | <del>20</del> |
| <del>21</del> | <del>22</del> | <b>23</b>     | <del>24</del> | <del>25</del> | <del>26</del> | <del>27</del> | <del>28</del> | <b>29</b>     | <del>30</del> |
| 31            | <del>32</del> | <del>33</del> | <del>34</del> | <del>35</del> | <del>36</del> | 37            | <del>38</del> | <del>39</del> | <del>40</del> |
| 41            | <del>42</del> | 43            | <del>44</del> | <del>45</del> | <del>46</del> | 47            | <del>48</del> | <del>49</del> | <del>50</del> |
| <del>51</del> | <del>52</del> | 53            | <del>54</del> | <del>55</del> | <del>56</del> | <del>57</del> | <del>58</del> | 59            | <del>60</del> |

The next number in the list that has not been eliminated is 11. There are no multiples of 11 left on our list, so there is nothing to eliminate.

The same is true for 13, 17, 19, 23, and 29.

# Sieve Up to 60, Continued

Use the Sieve of Eratosthenes to find all prime numbers up to 60.

|           |          |           |    |          |    |           |    |           |    |
|-----------|----------|-----------|----|----------|----|-----------|----|-----------|----|
| 1         | <b>2</b> | <b>3</b>  | 4  | <b>5</b> | 6  | <b>7</b>  | 8  | 9         | 10 |
| <b>11</b> | 12       | <b>13</b> | 14 | 15       | 16 | <b>17</b> | 18 | <b>19</b> | 20 |
| 21        | 22       | <b>23</b> | 24 | 25       | 26 | 27        | 28 | <b>29</b> | 30 |
| <b>31</b> | 32       | 33        | 34 | 35       | 36 | <b>37</b> | 38 | 39        | 40 |
| <b>41</b> | 42       | <b>43</b> | 44 | 45       | 46 | <b>47</b> | 48 | 49        | 50 |
| 51        | 52       | <b>53</b> | 54 | 55       | 56 | 57        | 58 | <b>59</b> | 60 |

The remaining numbers on the list are more than 30, so they won't have any multiples on our list. So they must all be prime numbers.

So prime numbers up to 60 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, and 59.